

Unit 4 - PM

MC - Questions Answered
on D2L

MC	Written	Total
/14	/21	/35

Knowledge/Application

- Determine the exact value of the following expression, $\frac{\sin \frac{2\pi}{3} + \sec \frac{\pi}{6}}{\tan \frac{11\pi}{6} \sin \frac{\pi}{6}}$.
- Given that $\sin A = \frac{4}{5}$, $A \in \left(0, \frac{\pi}{2}\right)$, and $\tan B = -\frac{12}{5}$, $B \in \left(\frac{\pi}{2}, \pi\right)$, determine the exact value of $\cos(A - B)$.

Application

- Use an appropriate compound angle formula to determine the exact value of $\sin \frac{-7\pi}{12}$.
- Julia is flying a kite at the end of a 40 m string. The string makes an angle of $\frac{\pi}{6}$ with the ground. The wind increases and the kite flies higher until the string makes an angle of $\frac{\pi}{4}$ with the ground. Determine an exact expression for the vertical distance that the kite moves between the two positions, **then evaluate** to a tenth of a meter. Draw a diagram to support your solution.

Communication

- Explain how you determine how many radians represents $7/8$ of a revolution. Then **state** the exact value.
- Alex finds that the SIN key on his calculator is not working. Use a co-function identity to determine another way he can evaluate $\sin\left(\frac{10\pi}{9}\right)$.

Thinking

- Solve $2\sin^2 x = 3\cos x + 3$, $x \in [0, 2\pi]$.
- Prove that $\frac{1 - \cos 2x + \sin 2x}{1 + \cos 2x + \sin 2x} = \tan x$